

Genome Pods for Clinical Practice



Elias Crum

vito.be

Outline

Why?



Genomic Data in Health Care?



Genomic Data in Solid Pods?



How Solid is helpful



Example clinical workflow introduction

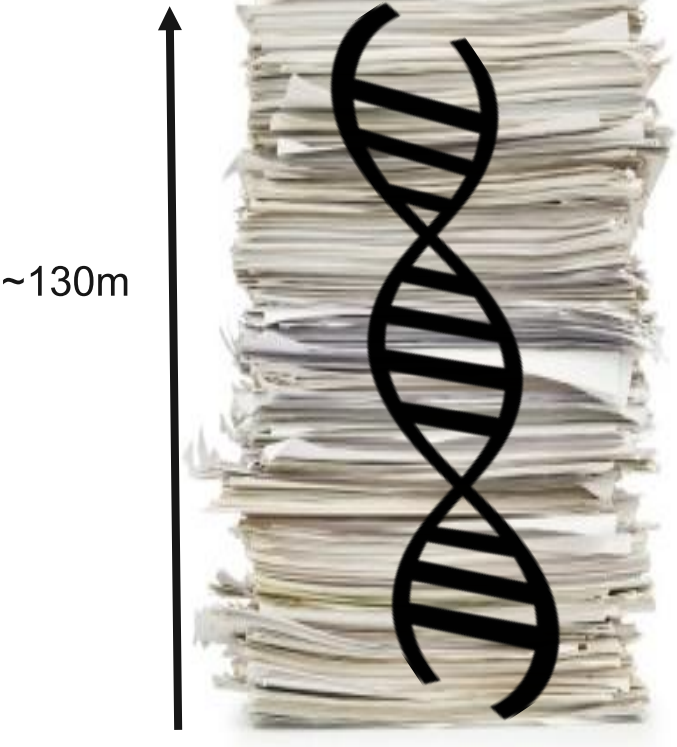
How?



“Demo” workflow of Genome Pods UI

Genome Data

Genome data is big...



~130m



Ghent Belfry

~91m

Other considerations

 Storage

 Privacy

 Governance

 Formatting

Printed

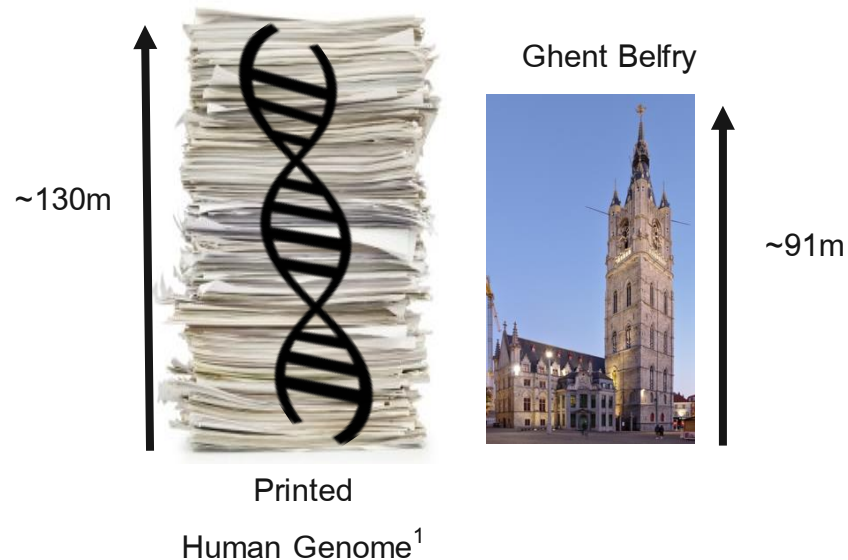
Human Genome¹

Genome Data: One Challenge “Solved”

Genome data is big...

Variant Call Format (VCF) file

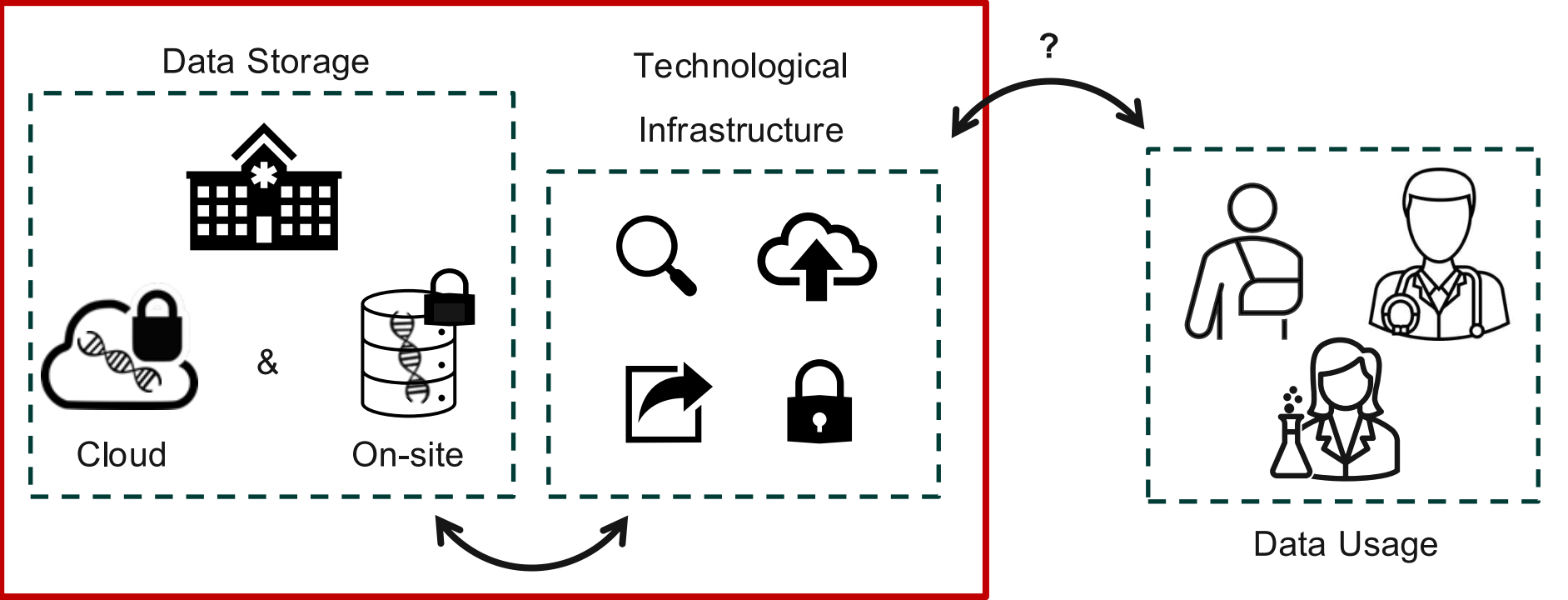
- Individual humans exhibit ~3 million mutations (~**125 MBs**)
- VCF file **records only those differences**



Other considerations: ??

Genome Data in Health Care

Institution-centric



Scaling Genome Data Usage

Research genomic data sharing initiatives:



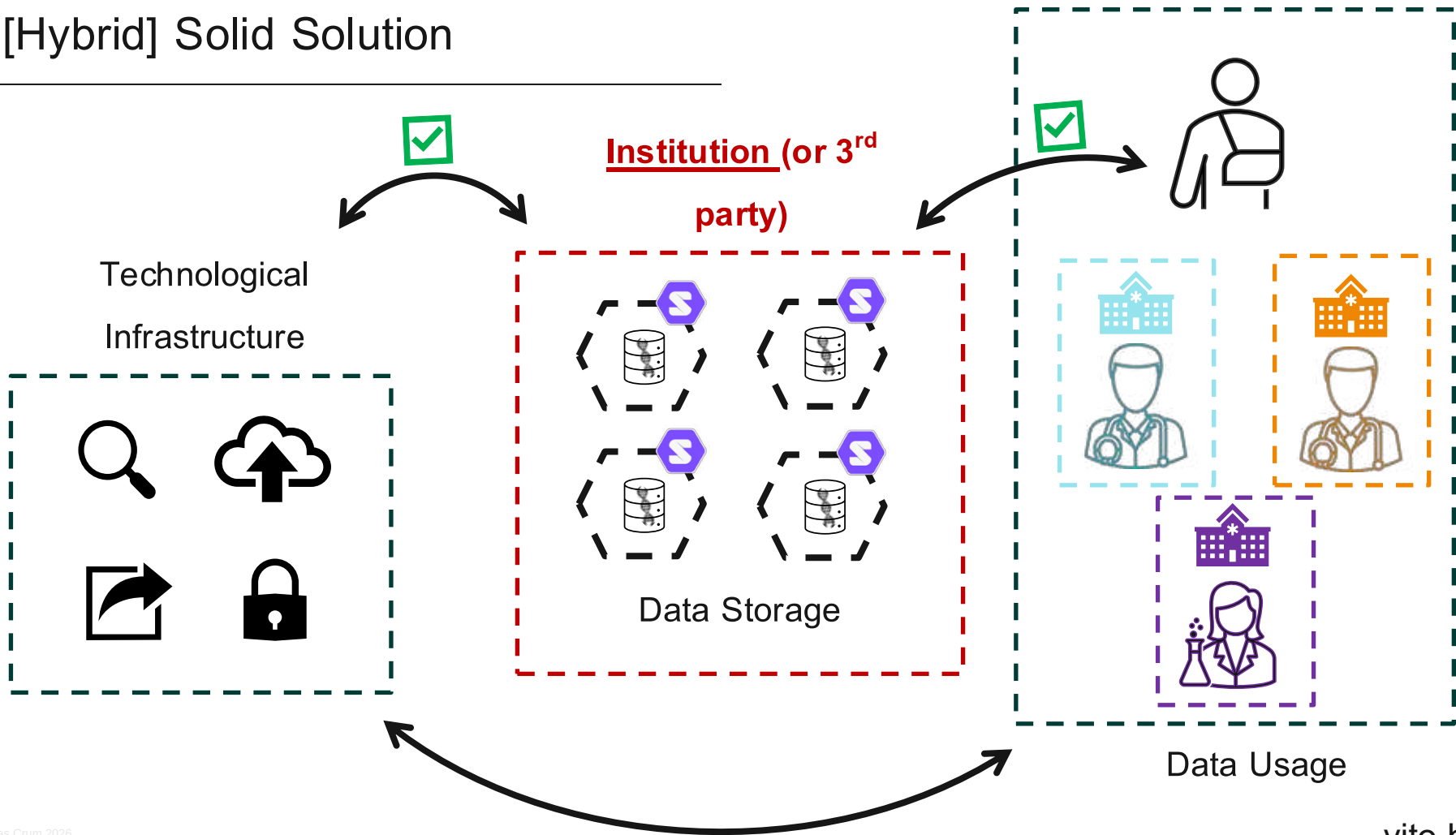
State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US) *EU Health
Data Spaces*



Not much...

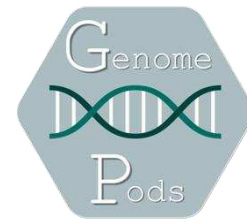
A [Hybrid] Solid Solution





- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

Disclaimer & Project Goals



The Genome Pods project is a **RESEARCH** project

It aims to develop a **proof-of-concept**

Framework and **Web Application** for:

- Genomic data storage in Solid Pods
- Granular privacy controls of genomic data
- Genomic data representation in RDF
- Data-linking potential
- Data querying potential (within and across pods)

Clinical Example

A patient is hospitalized with DVT

(DVT usually merits the perscription of Warfarin²)

The physician has just received a
pharmacogenomic screening

(produced with the patient's VCF file)

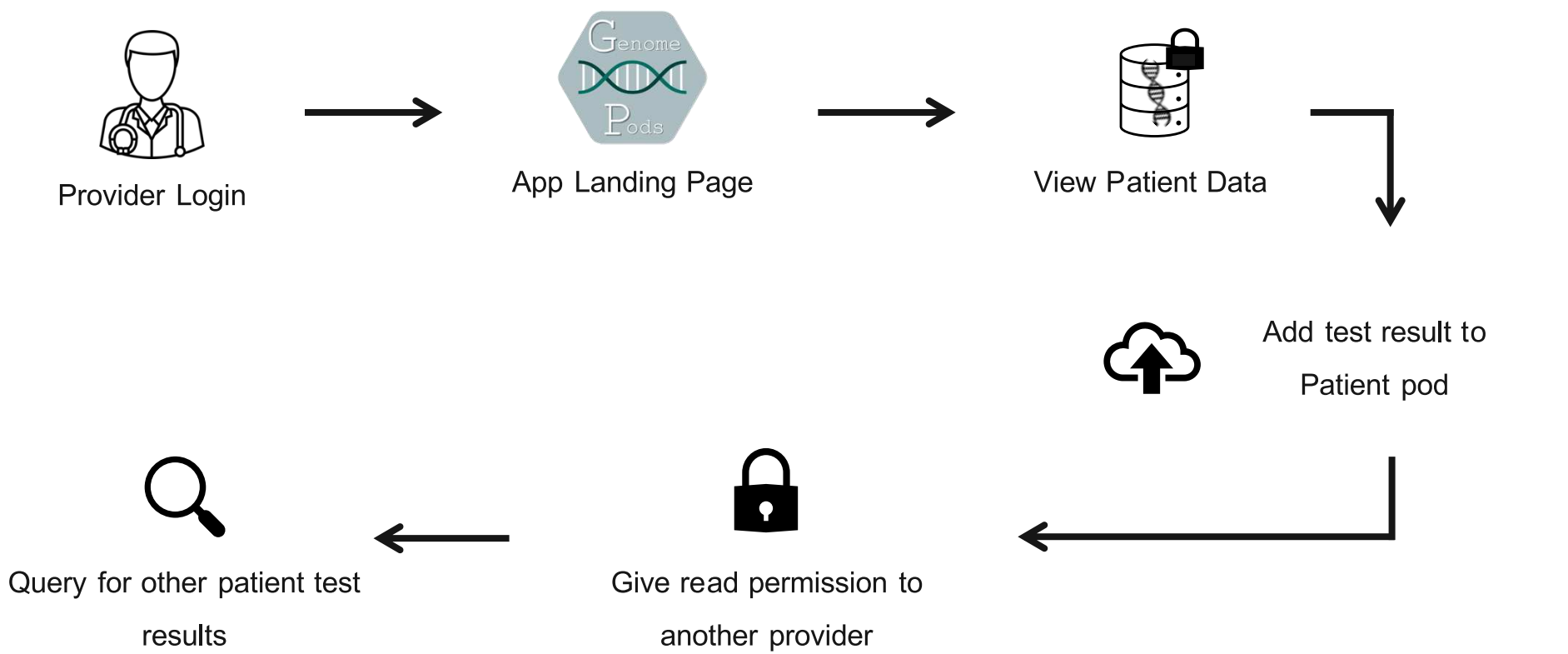
The physician uploads the test results to the
patient's Genome Pod

The physician makes the test results
visible to the patient's primary care
physician

(out of network)

The physician queries his other patients for
similar tests

Example Workflow: Physician



Figma example for workflow

[Open Here](#)

Future Directions

Basic Functionalities:



Genomic data storage in Solid pods



Data privacy management



Data Sharing via the web

Experimental research:



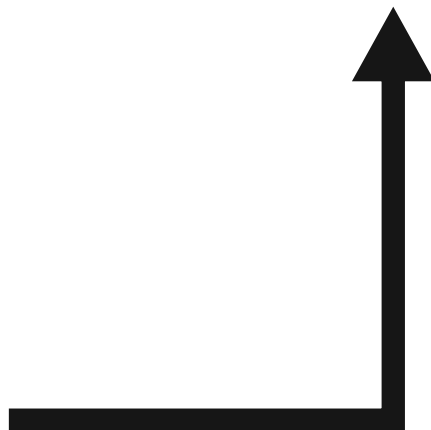
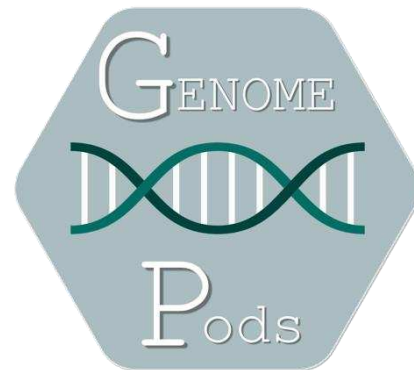
Represent genomic data as RDF?



Linking genomic and health data?



Single and Many pod Querying?



Thank you for listening!



Questions?



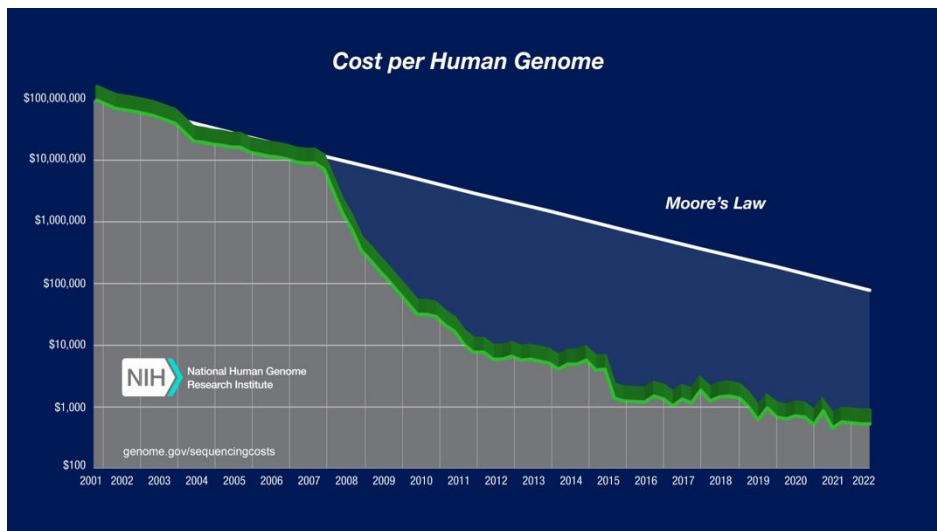
Part of the *PERsoNal Genome QUery IN clinical practice (PENGQUIN)* Ph.D. Project

Funding provided by VITO NV (UG_PhD_2303_contract)

vito.be

Genome Sequencing in Health Care

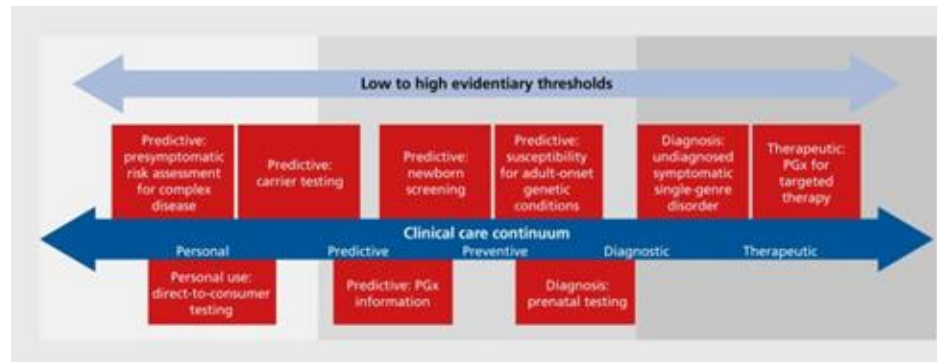
Sequence Generation Costs



Wetterstrand KA. Accessed 12/04/24.

Currently getting close to \$100 per genome⁴

Use-cases

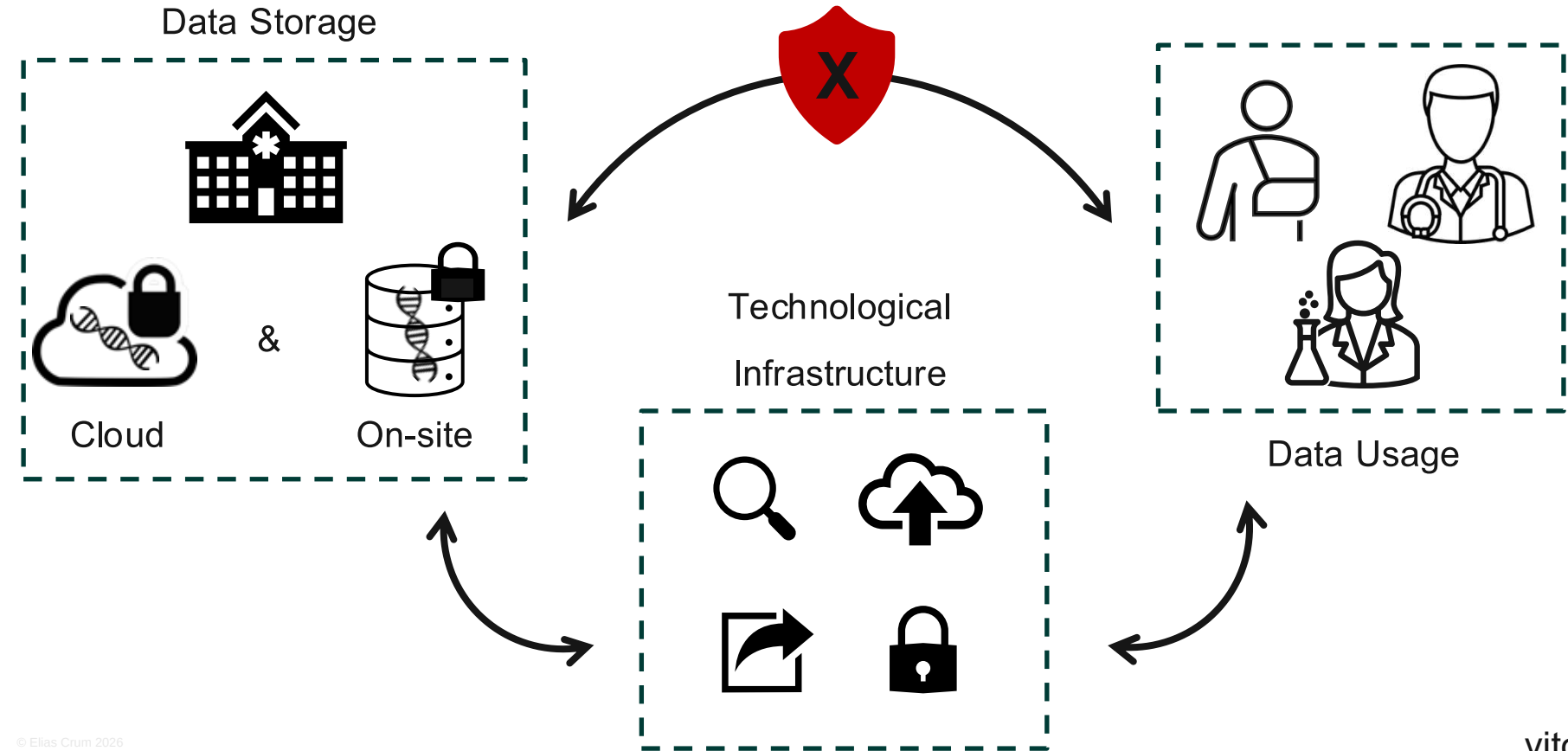


10.31887/DCNS.2016.18.3/jkrier

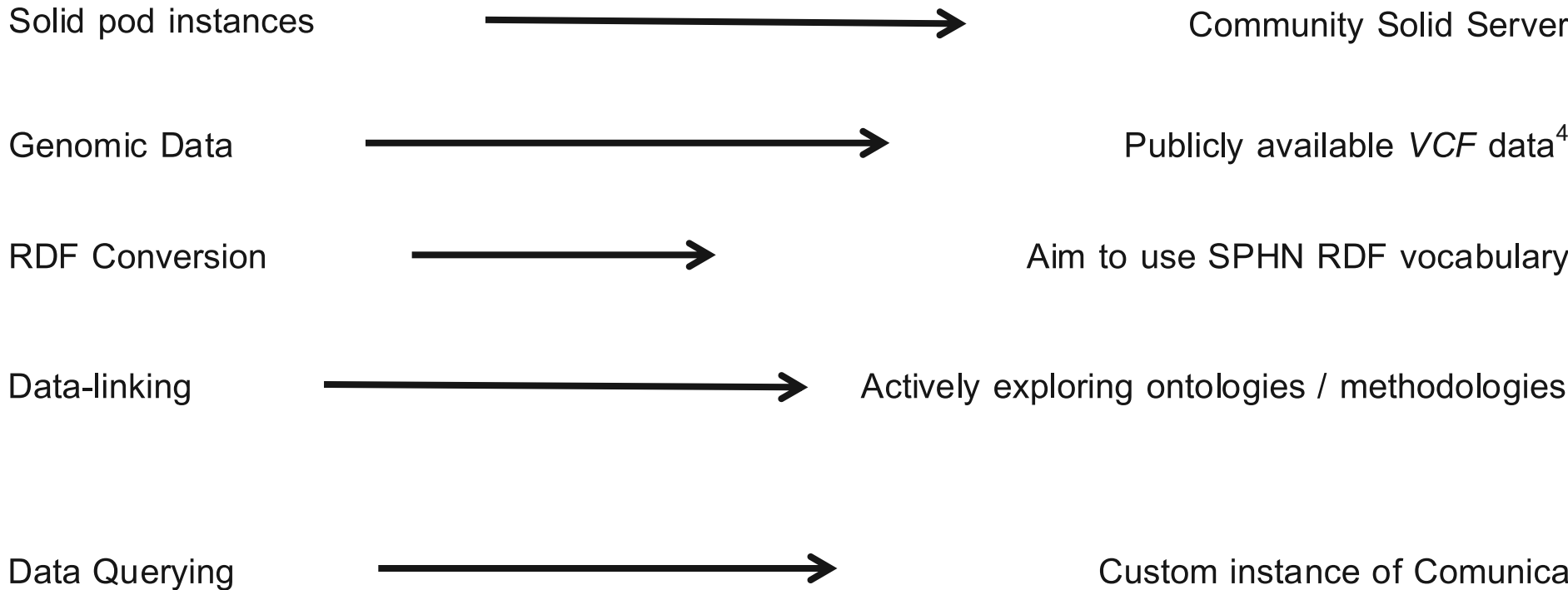
Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Genome Data in Health Care



Implementation Specifics



Sources

1. https://bio4.us/biotrends/human_genome_height.html
2. Tse, G. et al. Genotype-guided warfarin dosing vs. conventional dosing strategies: a systematic review and meta-analysis of randomized controlled trials. 2018.
<https://doi.org/10.1111%2Fbcp.13621>
3. Wetterstrand KA. DNA Sequencing Costs: Data from the NHGRI Genome Sequencing Program (GSP) www.genome.gov/sequencingcostsdata. Accessed 12/04/24.
4. <https://www.ultimagenomics.com/publications-and-resources?filter=Literature>